

MECH5200 Numerical Methods for PDEs : *Video 5:* *Deriving the Transverse Deflection of a String*

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Governing equation for a 1-D string deflection

- The deflection of a 1-D string.

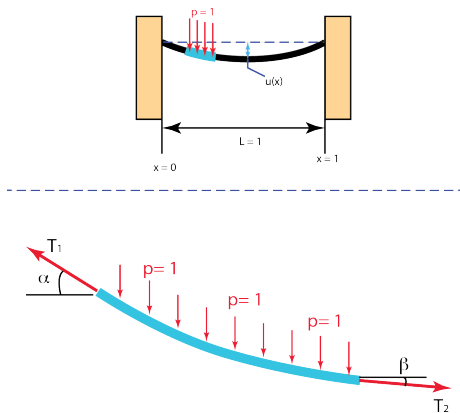


Figure: Deriving the String Equation: String deflection under distributed load, p

Governing equation for a 1-D string deflection

- Equilibrium in the x-direction (assuming small angles):

$$T_1 \cos(\alpha) = T_2 \cos(\beta) \simeq k\delta x \quad (1)$$

- Equilibrium in the y-direction:

$$T_1 \sin(\alpha) - T_2 \sin(\beta) - p\Delta x = 0 \quad (2)$$

Governing equation for a 1-D string deflection

- Divide the Equilibrium in y-direction by $T_1 \cos(\alpha) = T_2 \cos(\beta) \simeq k\delta x$

$$\frac{T_1 \sin(\alpha)}{T_1 \cos(\alpha)} - \frac{T_2 \sin(\beta)}{T_2 \cos(\beta)} = \frac{p\Delta x}{k\delta x}$$

$$\tan(\alpha) - \tan(\beta) = \frac{p\Delta x}{k\delta x}$$

Governing equation for a 1-D string deflection

- Recognizing, $\tan(\alpha)$ is the slope of the string at $x - \frac{\Delta x}{2}$, and $\tan(\beta)$ is the slope of the string at $x + \frac{\Delta x}{2}$:

$$\begin{aligned}\tan(\alpha) - \tan(\beta) &= \frac{f \Delta x}{k \delta x} \\ \frac{\partial u}{\partial x} \Big|_{x - \frac{\Delta x}{2}} - \frac{\partial u}{\partial x} \Big|_{x + \frac{\Delta x}{2}} &= \frac{p \Delta x}{k \delta x}\end{aligned}$$

- Recognizing: the left hand side is the difference in slopes at the end points of the string, and dividing both side of the equation by Δx :

$$\frac{\Delta \frac{\partial u}{\partial x}}{\Delta x} = \frac{p}{k \delta x} \quad (3)$$

Governing equation for a 1-D string deflection

- Taking the limit as $\Delta x \rightarrow 0$, we recover the derivative:

$$\frac{\partial^2 u}{\partial x^2} = \frac{p}{k\delta x} \quad (4)$$

- Setting $f = \frac{p}{k\delta x}$, we recover the Poisson equation:

$$\nabla^2 u = f \quad (5)$$

Governing equation for a 1-D string deflection

- The deflection of a string is captured by the Poisson Equation

$$\nabla^2 u = f \quad (6)$$